ALGORITHM ARENA

Iterated Prisoner's Dilemma

Axelrod's Classic 2-Player Game

Document A: Formal Problem Specification

Problem Identifier:	axelrod-ipd
Specification Level:	Document A (Pure Problem Definition)
Version:	1.0.0
Category:	Game Theory / Multi-Agent Decision Making
Difficulty:	Intermediate
Date:	August 2026

This document establishes the mathematical and domain-level specification of the Iterated Prisoner's Dilemma benchmark, independent of platform execution runtimes and tournament matchmaking implementations.

Contents

1 Metadata

1.1 Problem Name

Iterated Prisoner's Dilemma (Axelrod IPD)

1.2 Short Identifier

`axelrod-ipd`

1.3 Category

- ■ Game Theory
- ■ Multi-Agent Systems
- ■ Online Decision Making
- □ Continuous Control
- □ Combinatorial Optimization

1.4 Difficulty Level

- □ Beginner
- ■ Intermediate
- □ Advanced
- □ Research

1.5 Version History

Version	Date	Author	Changelog Summary
0.1.0	2026-08-19	Arena Core Team	Initial concept formulation and problem outline
1.0.0	2026-08-20	Arena Core Team	Complete standardized problem specification (Document A)

2 Problem Overview

2.1 Description

The *Prisoner's Dilemma* is the canonical non-zero-sum 2×2 symmetric game in modern mathematical game theory. Two agents interact by choosing simultaneously between two discrete actions: **Cooperation** (C) and **Defection** (D). In a single-shot interaction, defection strictly dominates cooperation for each individual agent regardless of the opponent's choice; however, mutual cooperation results in a strictly higher joint welfare than mutual defection.

In the **Iterated Prisoner's Dilemma (IPD)**, the stage game is repeated sequentially over a fixed horizon of $T = 200$ rounds. Because players maintain complete memory of past joint decisions, repetition unlocks complex reciprocal dynamics, enabling retaliation against defection, mutual trust building, punishment, and forgiveness.

2.2 Motivation

The Iterated Prisoner's Dilemma serves as the inaugural benchmark for Algorithm Arena because it exemplifies fundamental algorithmic principles:

1. **Tension Between Rationality and Welfare:** Captures the mathematical conflict between myopic personal payoff maximization and long-term reciprocal cooperation.

2. **Opponent-Dependent Dynamics:** The effectiveness of any decision rule cannot be evaluated in isolation; its performance is strictly coupled to the behavior and memory of the opponent.

3. **Clean Strategic Depth:** Despite minimal action space ($|\mathcal{A}| = 2$), optimal behavior requires state estimation, opponent profiling, and adaptive retaliation.

2.3 Learning Objectives

By analyzing and designing algorithms for this problem, researchers and competitors will master:

- The formalization of stage games versus repeated games with finite horizons.
- Nash equilibrium versus Subgame-Perfect Equilibrium (SPE) and the Folk Theorem.
- Designing reactive, memory-bounded, and stochastic policies in adversarial and cooperative multi-agent environments.

3 Formal Problem Definition

3.1 Agents and Roles

Each match consists of $N = 2$ symmetric autonomous agents:

$$\text{Agent 1} \quad \text{and} \quad \text{Agent 2}$$

Both agents possess identical action sets, payoff structures, and observation formats.

3.2 Horizon and Timing

- **Game Horizon:** Fixed finite duration of $T = 200$ sequential rounds: $t \in \{0, 1, \ldots, T-1\}$.
- **Decision Timing:** Discrete-time, synchronous lock-step execution. In each round t, both agents select their actions simultaneously without observing the opponent's current choice $a_t^{(\text{opp})}$.

3.3 Action Space

At each round t, agent $i \in \{1, 2\}$ selects an action $a_t^{(i)}$ from the discrete action set $\mathcal{A}$:

$$\mathcal{A} = \{C, D\} \equiv \{1, 0\}$$

where:

- $C = 1$: Cooperate
- $D = 0$: Defect

3.4 Payoff Matrix

The stage game payoffs for joint action pairs $(a^{(1)}, a^{(2)}) \in \mathcal{A} \times \mathcal{A}$ are defined by the canonical Axelrod payoff matrix:

		Agent 2	
		Cooperate (C)	**Defect (D)**
Agent 1	**Cooperate (C)**	$(R, R) = (3, 3)$	$(S, T) = (0, 5)$
	Defect (D)	$(T, S) = (5, 0)$	$(P, P) = (1, 1)$

The payoff parameters satisfy the two mandatory mathematical conditions of the Prisoner's Dilemma:

1. **Strict Incentive Order:**
$$T > R > P > S \quad \iff \quad 5 > 3 > 1 > 0$$
where T represents Temptation, R represents Reward, P represents Punishment, and S represents Sucker's payoff.

2. **Social Welfare Efficiency:**
$$2R > T + S \quad \iff \quad 2(3) > 5 + 0 \quad (6 > 5)$$
ensuring that sustained mutual cooperation yields strictly higher aggregate payoff than alternating exploitation.

3.5 State and Observation Space

Let h_t represent the public history of joint actions up to round $t - 1$:

$$h_t = \left((a_0^{(1)}, a_0^{(2)}), (a_1^{(1)}, a_1^{(2)}), \ldots, (a_{t-1}^{(1)}, a_{t-1}^{(2)})\right) \in (\mathcal{A} \times \mathcal{A})^t$$

At $t = 0$, the history is empty: $h_0 = \emptyset$.

At round t, the observation provided to agent i is:

$$o_t^{(i)} = \left(t,\ \mathbf{a}_{0:t-1}^{(i)},\ \mathbf{a}_{0:t-1}^{(j)}\right)$$

where $\mathbf{a}_{0:t-1}^{(i)}$ is the sequence of the agent's own past actions and $\mathbf{a}_{0:t-1}^{(j)}$ is the sequence of the opponent's past actions.

3.6 Transition Function

The environment state transitions deterministically by appending the newly selected joint actions:

$$h_{t+1} = h_t \cup \{(a_t^{(1)}, a_t^{(2)})\}$$

3.7 Reward Function and Total Match Utility

The stage reward $r_t^{(i)}$ for agent i against opponent j is:

$$r_t^{(i)} = R_i(a_t^{(i)}, a_t^{(j)}) = \begin{cases} R = 3, & \text{if } a_t^{(i)} = 1 \text{ and } a_t^{(j)} = 1 \\ S = 0, & \text{if } a_t^{(i)} = 1 \text{ and } a_t^{(j)} = 0 \\ T = 5, & \text{if } a_t^{(i)} = 0 \text{ and } a_t^{(j)} = 1 \\ P = 1, & \text{if } a_t^{(i)} = 0 \text{ and } a_t^{(j)} = 0 \end{cases}$$

The cumulative utility for agent i in a match of length $T = 200$ is:

$$U_i(h_T) = \sum_{t=0}^{T-1} r_t^{(i)}(a_t^{(i)}, a_t^{(j)})$$

4 Mathematical Model

4.1 Notation Summary

Symbol	Domain	Meaning
N	$\{2\}$	Number of agents in a match
t	$\{0, 1, \ldots, T-1\}$	Round index
T	$\mathbb{N}$ (200)	Total rounds per match
$\mathcal{A}$	$\{0, 1\} \equiv \{D, C\}$	Discrete action space
$a_t^{(i)}$	$\mathcal{A}$	Action chosen by agent i at round t
h_t	$(\mathcal{A} \times \mathcal{A})^t$	History sequence of joint actions up to round $t-1$
$r_t^{(i)}$	$\{0, 1, 3, 5\}$	Stage payoff received by agent i at round t
U_i	$[0, 5T] \equiv [0, 1000]$	Cumulative match payoff of agent i
π_i	$\mathcal{H} \to \mathcal{P}(\mathcal{A})$	Decision policy mapping observable history to action

4.2 Theoretical Properties

1. **Single-Stage Dominance:** Defection strictly dominates cooperation:

$$\forall a^{(2)} \in \{0, 1\}, \quad R_1(0, a^{(2)}) > R_1(1, a^{(2)})$$

   The unique pure-strategy Nash equilibrium of the single-shot game is $(D, D) = (0, 0)$, producing $(1, 1)$.

2. **Folk Theorem in Repeated Settings:** In an infinite or unknown horizon game with discount factor $\delta \geq \frac{T-R}{T-P} = 0.5$, mutual cooperation is supported as a Subgame-Perfect Equilibrium through trigger retaliation. In known fixed finite games of length T, backward induction predicts mutual defection in theoretical self-play; however, in empirical multi-agent tournaments with boundedly rational opponents, reciprocal cooperative strategies dominate.

5 Concrete Match Examples

5.1 Example 1: Mutual Cooperation (Tit-for-Tat vs. Pavlov)

Round t	Agent 1 (TFT)	Agent 2 (WSLS)	Payoff 1	Payoff 2	Outcome
0	C (1)	C (1)	3	3	Mutual Cooperation
1	C (1)	C (1)	3	3	Mutual Cooperation
2	C (1)	C (1)	3	3	Mutual Cooperation
3	C (1)	C (1)	3	3	Mutual Cooperation
4	C (1)	C (1)	3	3	Mutual Cooperation
Total (5 rds)			**15**	**15**	**Tie (Social Welfare Optimal)**

5.2 Example 2: Exploitation Retaliation (Tit-for-Tat vs. Always Defect)

Round t	**Agent 1 (TFT)**	**Agent 2 (ALLD)**	**Payoff 1**	**Payoff 2**	**Outcome**
0	C (1)	D (0)	0	5	Agent 2 Exploits Sucker
1	D (0)	D (0)	1	1	Agent 1 Retaliates
2	D (0)	D (0)	1	1	Mutual Defection
3	D (0)	D (0)	1	1	Mutual Defection
4	D (0)	D (0)	1	1	Mutual Defection
Total (5 rds)			**4**	**9**	**Agent 2 Wins Match**

6 References

1. Axelrod, R. (1980). *Effective Choice in the Prisoner's Dilemma.* Journal of Conflict Resolution, 24(1), 3-25.

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